Welcome to **week one**. This week is all about building **intuition for derivatives**.
What's a **derivative**? Think **velocity**. Just like a **speedometer** tells you how fast position changes, a derivative tells you **how fast one quantity changes with respect to another**.
We'll start with this **speedometer example**, then explore **fundamental mathematical functions**:

- **Constant functions**
- **Linear functions**
- **Quadratic polynomials**
- **Exponential functions**
- **Logarithmic functions**

You'll learn how to compute the **derivatives of each of these functions**.
To handle more complex functions, we'll introduce key **derivative rules**:

- **Sum rule**
- **Product rule**
- **Chain rule**
- **Multiplication by scalars**

Now, **why are derivatives so important in machine learning?**
Because they allow us to **optimize functions**—that is, **find maximum and minimum values**.
In **machine learning**, finding the **best model** means:

- Defining a **loss function**
- **Minimizing** that loss function using **derivatives**

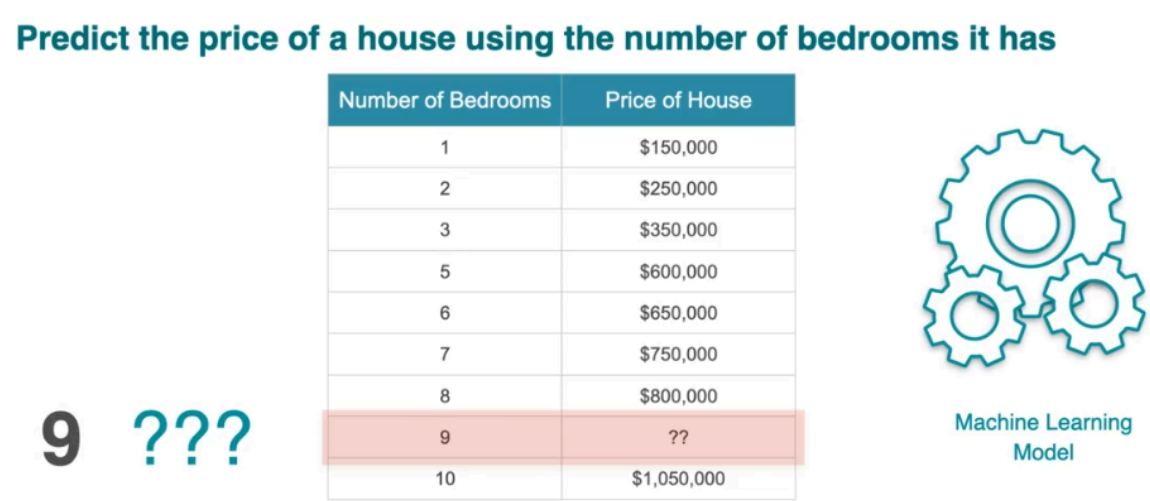
We'll look at several **minimization examples**, starting with a **temperature-based example**, and then move on to two **crucial loss functions in machine learning**:

- **Squared loss**
- **Log loss**

These concepts form the **foundation of optimization and learning algorithms** 🚀

I. Machine Learning Motivation

Problem to resolve:
We have 10 houses with bedrooms from 1 to 10. The 9th house doesn't have a price. Lets build a machine learning model to predict the house's price value.



Number of Bedrooms	Price of House
1	$150,000
2	$250,000
3	$350,000
5	$600,000
6	$650,000
7	$750,000
8	$800,000
9	??
10	$1,050,000

To make this easier lets plot the data
Y = Price of houses
X = Number of bedrooms. Houses are presented by dots

The idea of machine learning model is that the model will be trained on those data as inputs. The model represents under-the-hood (engine) of system of prediction.

I this example, the model is the line that goes as close as possible the all the point. The idea is the model is adjusted more and more to fit the most possible data point and make best possible prediction.

The 9th house is predicted to have price of 950.000

The previous problem is called the linear regression problem.

Imagine you travel to a distant planet and meet some alien life. You can interact with them, but you **don't understand their language**. Still, you notice something interesting.

One alien says **"aack, aack, aack"**, and you somehow know this alien is **happy**.
Another alien says **"beep, beep"**, and you know this one is **sad**.

You collect more observations. An alien says **"aack, beep, aack"** and is **happy**.
Another says **"aack, beep, beep, beep"** and is **sad**.

The goal here is **not to learn the alien language**—that might be far too complicated. The goal is much simpler and more practical: **predict whether an alien is happy or sad**.

So let's organize the **data set**. We have four sentences:

- **aack aack aack**
- **beep beep**
- **aack beep aack**
- **aack beep beep beep**

For each sentence, we extract **simple numerical features**:

- **Number of times "aack" appears**
- **Number of times "beep" appears**

And we also record the **alien's mood**: **happy or sad**.

Now the problem becomes clear.
Given a pair of numbers—**(count of aack, count of beep)**—we want to predict a **category**: **happy or sad**.

This is a **classification problem**: using measured features to assign an input to one of a **finite set of labels**.

Why? Because **given any sentence**, the goal is to **classify it as happy or sad**. Since we're assigning an **emotional label** to a piece of text, this task is called **sentiment analysis**.

Machine Learning Motivation

Sentence	Aack	Beep	Mood
Aack aack aack!	3	0	😊
Beep beep!	0	2	😔
Aack beep aack!	2	1	😊
Aack beep beep beep!	1	3	😔

A classification problem

Sentiment analysis

And things become much clearer when we **visualize the data**.

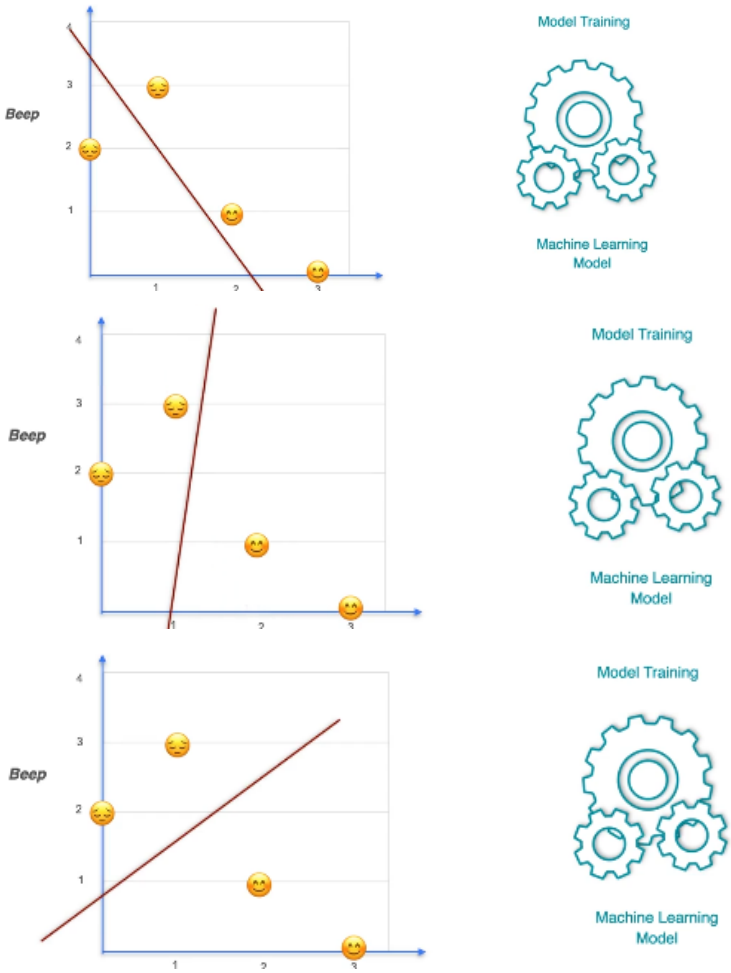
So let's make a **plot**. For each sentence:

- One axis represents the **number of times "aack" appears**
- The other axis represents the **number of times "beep" appears**

The idea of the **model** is to **separate the happy points from the sad points**.

Imagine we train a model, and after enough learning, it reaches an **ideal point**. This model effectively **splits the plane into two regions**:

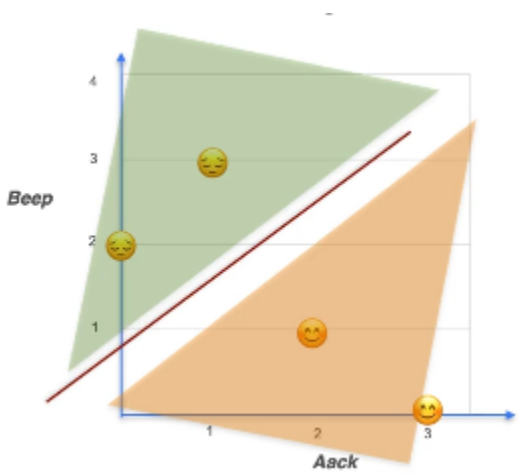
- **Sad region** in **green**
- **Happy region** in **orange**



Now, when a **new sentence** comes in, we count the “aack”s and “beep”s, plot it as a point, and see **which region it falls into**.

- If it falls in the **orange region**, the model predicts **happy**
- If it falls in the **green region**, the model predicts **sad**

The model may make mistakes, but for the most part it will do well, at least based on how it did on this data set.



It turns out there’s a lot of **mathematics driving the training process**. Key concepts include:

- **Gradients** and **derivatives**
- **Optimization**
- **Loss and cost functions**
- **Gradient descent**
  …and many more.

In this course, you’ll learn **how model training and optimization work at their core using mathematics**.

In our earlier example, we used **linear regression and classification models** to make the ideas concrete. But the techniques and concepts you learn here are **general** and can be applied to a wide range of machine learning algorithms, including **neural networks**.

II. Motivation to derivates-Part1

When I think of a **derivative**, the first thing that comes to mind is **velocity**.

Imagine you're driving a car and you travel **100 kilometers in one hour**. That tells you your **average velocity** was **100 km/h**. But in reality, your speed probably wasn't constant. Maybe you **started fast**, then **slowed down**, **stopped**, sped up again, and maybe even **went in reverse** for a moment.

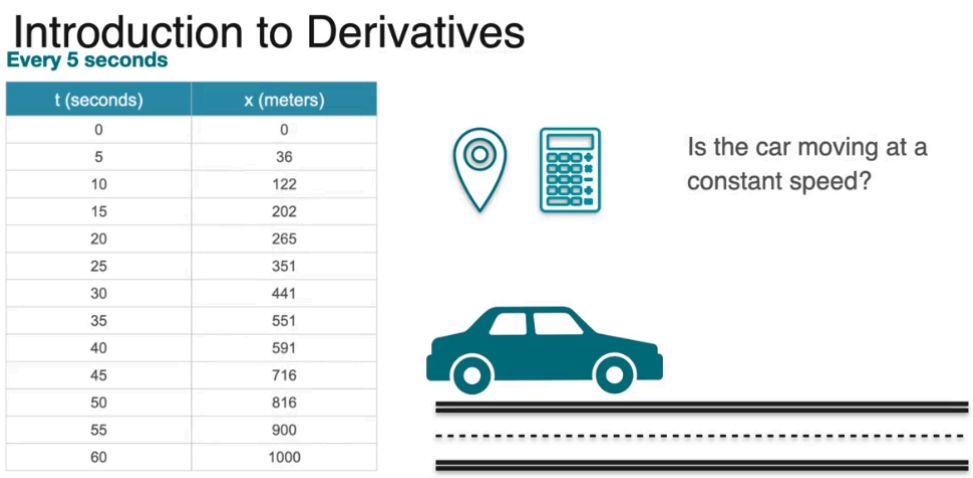
So the real question is: **at a given instant, what is your velocity right now?**
That's called **instantaneous velocity**.

And that is **exactly what a derivative is**.

A **derivative** is the **instantaneous rate of change of a function**.
In this example:

- The **function** is **distance**
- The **derivative** of distance is **velocity**

Let's add some numbers, the table resumes the distance every 5 seconds.



t (seconds)	x (meters)
0	0
5	36
10	122
15	202
20	265
25	351
30	441
35	551
40	591
45	716
50	816
55	900
60	1000

We can see that the car doesn't run at the same speed. Speed is not constant.

What is the average velocity of the car between 10s and 15s ? = 202-105 / 5s

<u>Lets plot the graph</u>

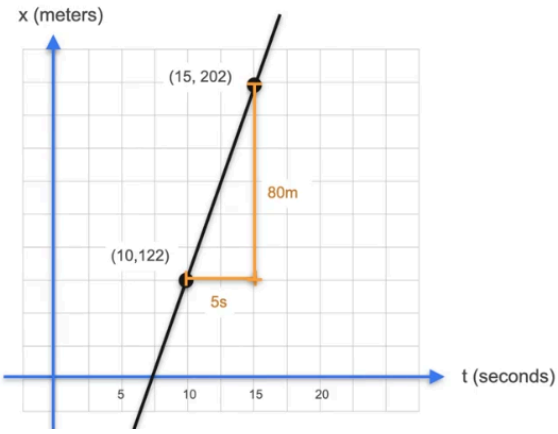
The **average velocity** between **10 and 15 seconds** is the **slope of the line** that connects the two corresponding points on the graph.

Why does this work? Because **velocity is distance over time**. And that's exactly the same idea as **slope**, which is **rise over run**.

- The **rise** is the **change in distance**
- The **run** is the **change in time**

Since **distance is on the vertical axis** and **time is on the horizontal axis**, the slope of the line between two points represents the **average velocity** over that time interval.

Calculating the Slope



$$slope = \frac{rise}{run}$$

$$slope = \frac{change\ in\ distance(\Delta x)}{change\ in\ time(\Delta t)}$$

$$slope = \frac{x(15) - x(10)}{t(15) - t(10)}$$

$$slope = \frac{202m - 122m}{15s - 10s}$$

$$slope = \frac{80m}{5s} = 16m/s$$

What is the velocity at 12.5 s ?

Recall that **slope = change in distance / change in time**, and this slope represents the **average velocity over the interval**.

So we compute:

- **Distance at 13 seconds:** 170
- **Distance at 12 seconds:** 155
- **Change in time:** 1 second

The slope is (170−155)/(13−12)=15 m/s

This **15 meters per second** is the **average velocity** over that interval—and it's a **much better estimate** of the **instantaneous velocity at t=12.5t = 12.5t=12.5**.

Still, it's not exact. So what do we do?
We take **finer and finer time intervals**. And as the interval gets **smaller**, the estimate gets **closer and closer** to the **true instantaneous velocity**. **That leads to derivative which is coming next.**

Quiz : Slope of a Line 2

t (s)	x (m)
10	122
11	138
12	155
13	170
14	186
15	202
16	218
17	234
18	250
19	265
20	265

Can you find a good estimate of the velocity of the car at time t = 12.5 seconds using this data?

$$slope = \frac{\Delta x}{\Delta t}$$

$$slope = \frac{x(13) - x(12)}{13s - 12s}$$

$$slope = \frac{170m - 155m}{1s}$$

$$slope = 15m/s$$

III. Derivatives and Tangents

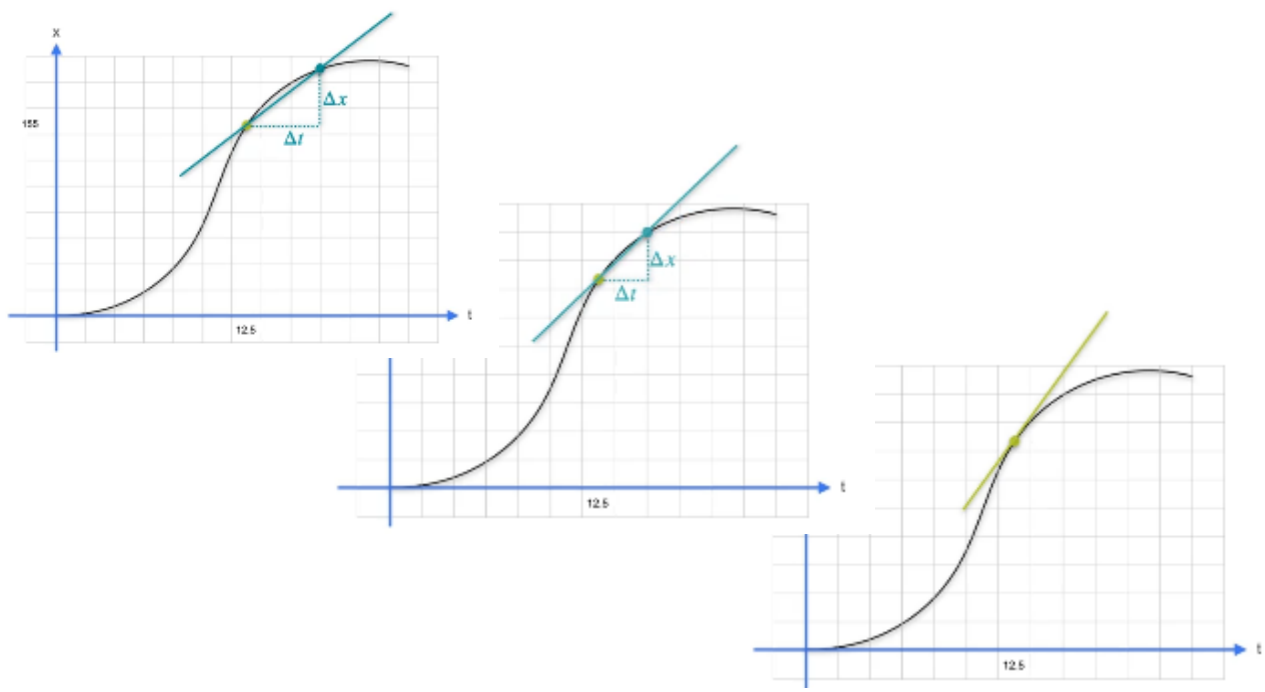
You have the curve of the changing of distance in function of changing in time.

You can estimate the velocity at 12.5, using two points.

This two point keeps getting smaller and smaller, till it get closer and closer till reach limit. At this point the line becomes tangent for the curve at t=12.5s and delta become d.

This **measure of how fast the distance is changing with respect to time** is called the ***instantaneous rate of change*** and **it is the slope of that tangent line**. ***This is the instantaneous rate of change*** that **is called a derivative**. **Keep in mind that the derivatives of a function at a point is the slope of the tangent.**

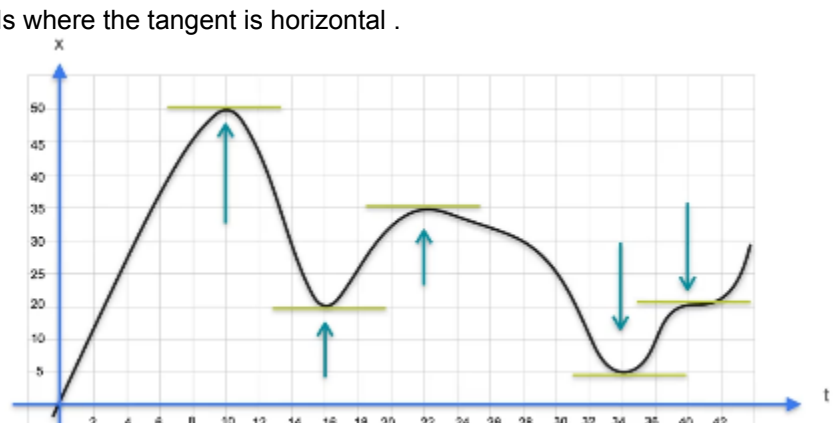
$$\frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \quad \frac{\Delta x}{\Delta t} \longrightarrow \frac{dx}{dt}$$



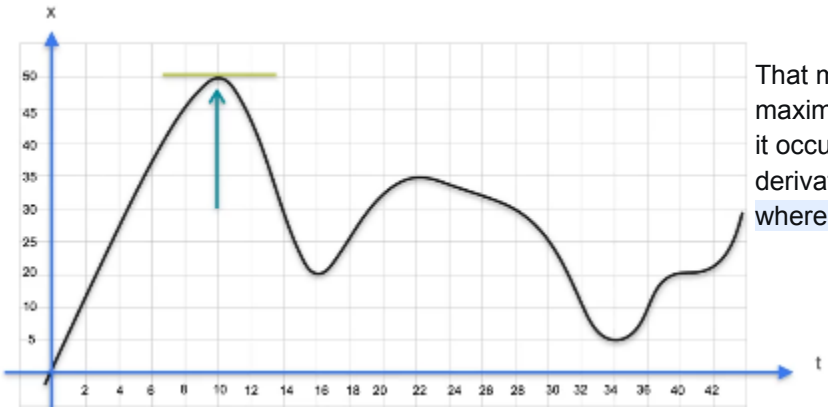
IV. Slopes, maxima and minima

Where was the point where velocity was 0 ?

Is where the tangent is horizontal .



When the car was the farthest from it's starting point ?



That means if you want to find the maximum or the minimum in a function, it occurs at one of the points where the derivative is zero. Or in other words, where the tangent line is horizontal.

The slope of a tangent is the value of the derivative of a given function at a specific point.

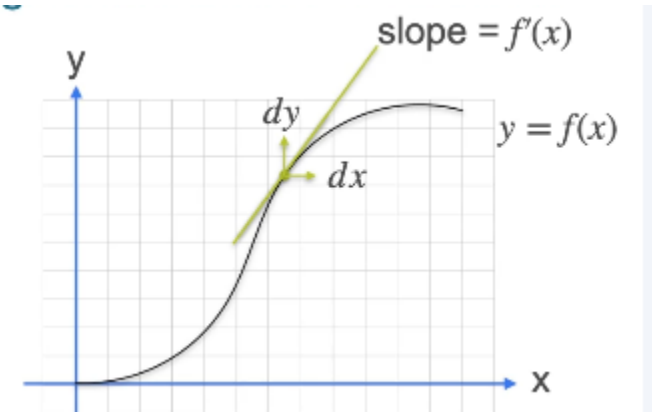
V. Derivatives and their notation

$$y = f(x)$$

Derivative of f is expressed as:

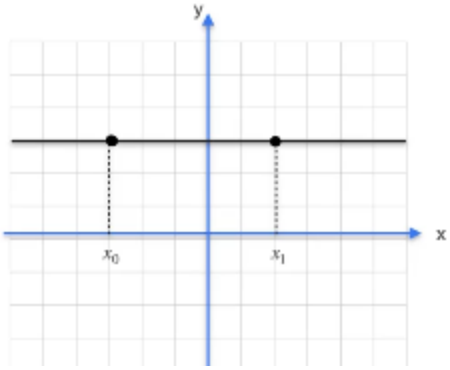
$f'(x)$ **Lagrange's notation**

$\frac{dy}{dx} = \frac{d}{dx} f(x)$ **Leibniz's notation**



VI. Some common derivatives - Lines

VI.1 Derivatives of a constant

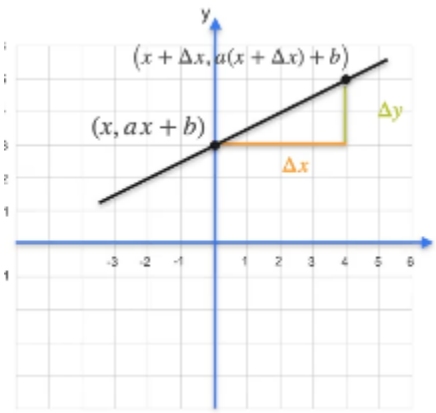


$$y = f(x) = c$$

Slope?

$$\frac{\Delta y}{\Delta x} = \frac{c - c}{x_1 - x_0} = 0 \rightarrow \boxed{f'(x) = 0}$$

VI.2 Derivatives of a line



$$f(x) = ax + b \Rightarrow f'(x) = a$$

$$\frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}} = a$$

$$\frac{df}{dx} \boxed{\frac{\Delta y}{\Delta x}} = \frac{a(\cancel{x} + \Delta x) + \cancel{b} - (\cancel{ax} + \cancel{b})}{\Delta x}$$

$$= a\frac{\cancel{\Delta x}}{\cancel{\Delta x}} = \boxed{a}$$

VI.2 Some common Derivatives - Quadratics, cubic

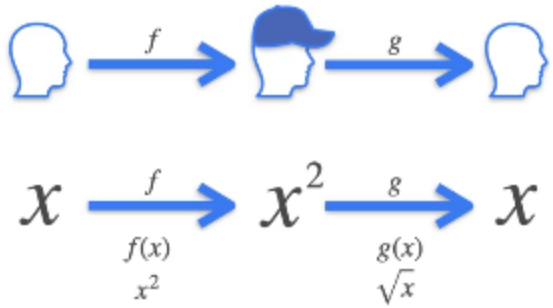
$$f(x) = x^2 \Rightarrow f'(x) = 2x$$

$$f(x) = x^3 \Rightarrow f'(x) = 3x^2$$

VII. The inverse function and its derivative

Inverse Function

What's an inverse?



$g(x)$ and $f(x)$ are inverses

$$g(x) = f^{-1}(x)$$

$$g(f(x)) = x$$

$$\sqrt{x^2} = x \quad \text{for } x > 0$$

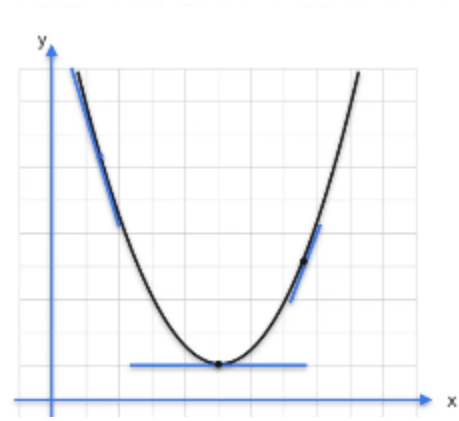
g(y) is the inverse of f(x) so

$$g'(y) = \frac{1}{f'(x)}$$

trigonometric functions are cos(x) and sin(x)

$$f(x) = \sin(x) \Rightarrow f'(x) = \cos(x)$$

$$g(x) = \cos(x) \Rightarrow g'(x) = -\sin(x)$$



VIII. Existence of the derivative

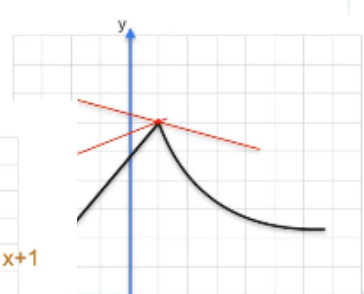
A function is **differentiable** at a point if its **derivative exists at that point**.

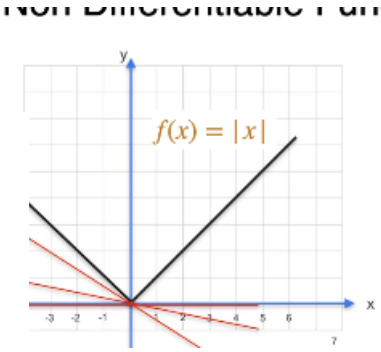
- A function is **differentiable at a point** if its **derivative exists at that point**.
- **Geometric meaning**: a function is differentiable at a point if you can draw a **unique, well-defined tangent line** at that point.

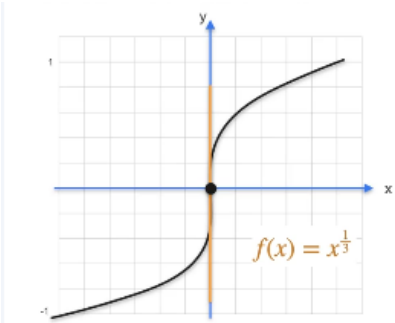
- A function is **differentiable on an interval** if the derivative exists **at every point in that interval**.
- Functions that **do not have a derivative at one or more points** are called **non-differentiable functions**.

A function is **not differentiable at a point** if:

- There is a **corner**
- There is a **cusp**
- **Jump discontinuity**
- **Function with vertical tangent**







$$\frac{3}{5} \begin{array}{l} \longleftarrow \text{numerator} \\ \longleftarrow \text{denominator} \end{array}$$

IX. Properties of the derivative

IX.1. Multiplication by scalars

When the function is multiplied by c, the derivative is going to be multiplied by c also.

IX.2. The sum rule

$$\boxed{\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)}$$

IX.3. The product rule

$$f'(t) = g'(t)h(t) + g(t)h'(t)$$

IX.3. The chain rule

$$\frac{d}{dt} f(g(h(t))) = \frac{df}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{dt}$$

$$= f'(g(h(t))) \cdot g'(h(t)) \cdot h'(t)$$

Note : if we want to compare two functions greater than or smaller than, we cant depend on the derivative(slope) because slope represents only the rate of how a particular function changes, we need the function itselves for the comparison.

when we calculate the derivative and replace x with a value f'(x = 5), the result is the slope value of the tangent at this point of f(x) curve

X. Introduction to optimization

X.1 minima and maxima

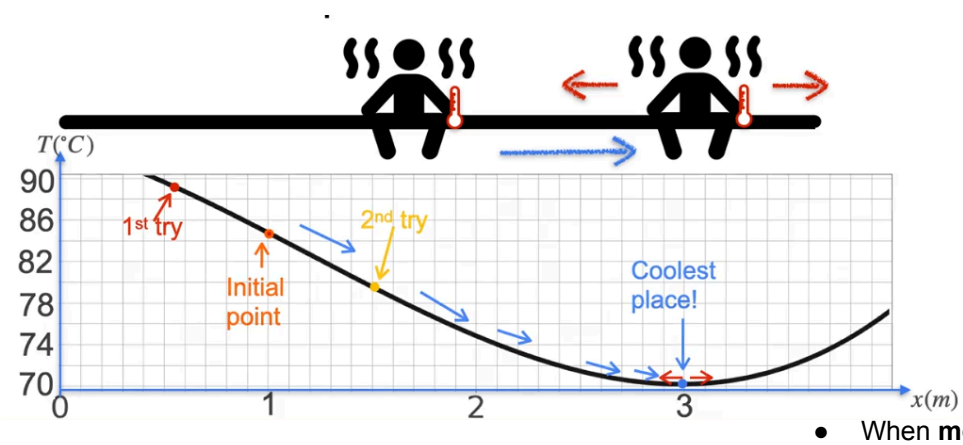
By now, you have **many tools to work with derivatives**. So the natural question is: **what are they actually useful for?** Beyond measuring rates of change, **why do derivatives matter in machine learning?**

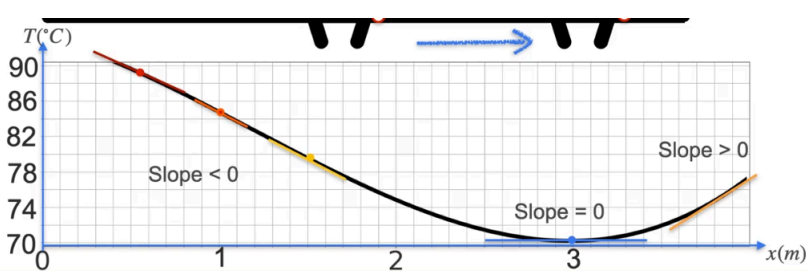
The main reason is **optimization**.
Optimization means finding the **maximum or minimum** of a function.

This is crucial in **machine learning** because the goal is to find the **model that best fits the data**. To do that, we define an **error (or loss) function** that measures **how far our model is from the ideal one**.

<u>Example</u> 👍

You are sitting in a bunch, where you are sitting getting hotter, you go right and left to see if there is more moderate temperature to be sitting at. Eventually, you reach a point where **any direction you move makes the temperature increase**. '3'





- When **moving to the right**, the **slope is negative**
- When **moving to the left**, the **slope is positive**
- At the **coldest point**, where **any movement makes things hotter**, the **slope is zero**

This reveals a key property of **maximum and minimum points**: 👉 **At these points, the slope (derivative) is zero.**

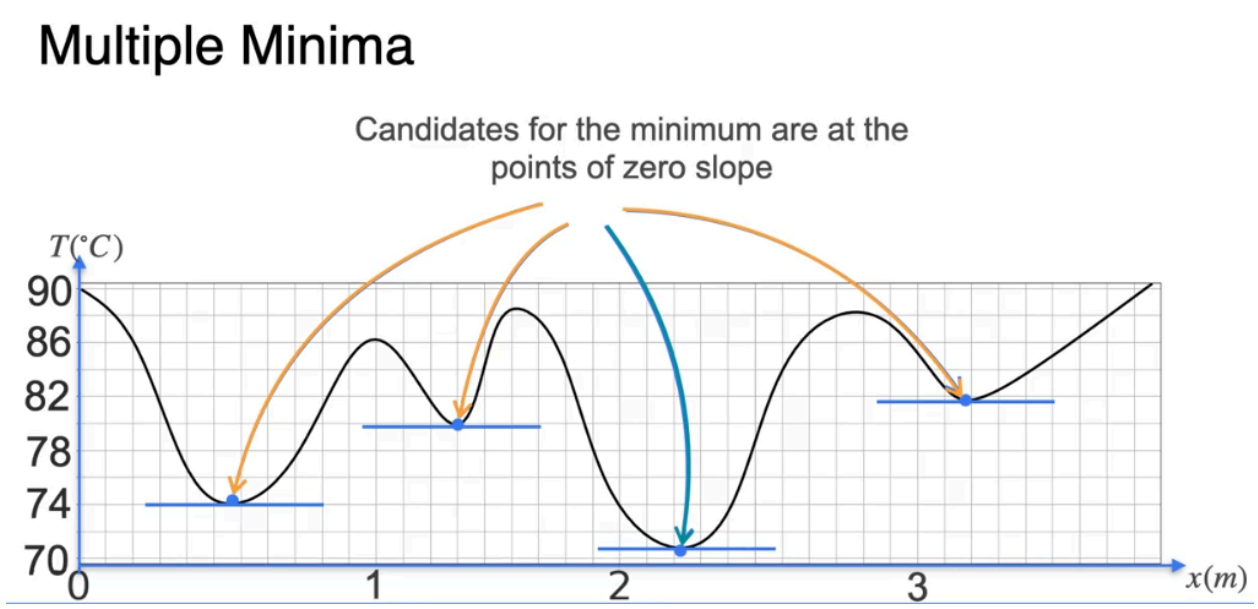
Imagine a more **complicated sauna**. The **coldest spot** is somewhere **to the right of 2**, but along the way there are **many points where the derivative is zero**. These points are **flat**, but not all of them are the coldest.

So what do we do?
You have to **check all of them** and see **which one is actually the coldest**. The good news is that calculus already helps you a lot: it **reduces the problem to a small set of candidate points**.

- Points where the **derivative is zero** are called **local minima** (or local maxima)
- The **coldest point overall** is the **global minimum**

When you want to **optimize a function**—whether **minimizing or maximizing**—and the function is **differentiable everywhere**, there's one key rule:

👉 **The only candidates for maxima and minima are the points where the derivative is zero.**

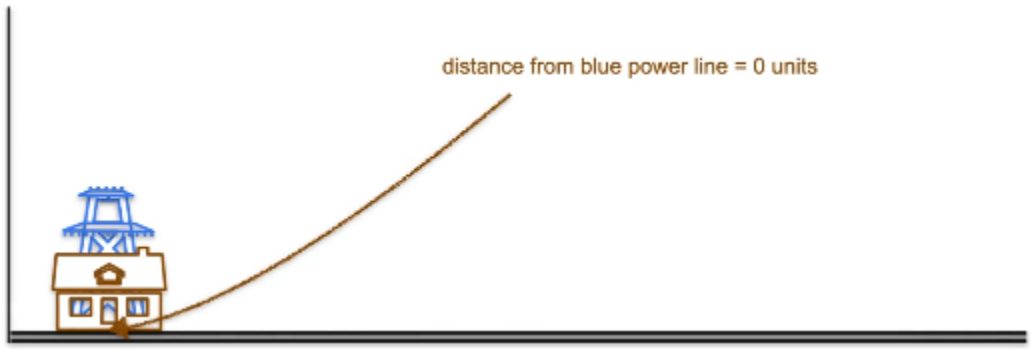


X.2. Optimization of squared loss - The one powerline problem

The idea is to **choose the best location to build a house** so that the **cost of connecting it to several power lines is minimized**. This example matters because the function we're minimizing is very similar to the **squared error**, one of the **most important loss functions in machine learning**. It's used in **linear regression** and in **certain neural networks**.

So the question becomes: **where should you place the house to minimize this total cost?**

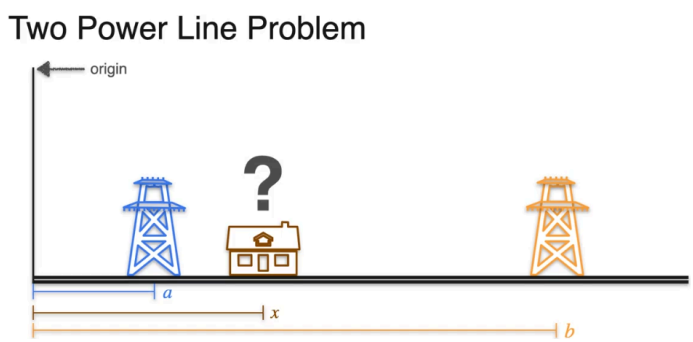
. If there's only one power line, the answer is obvious: you build the house right next to it. The distance is zero, and the cost is 0^2 = 0.

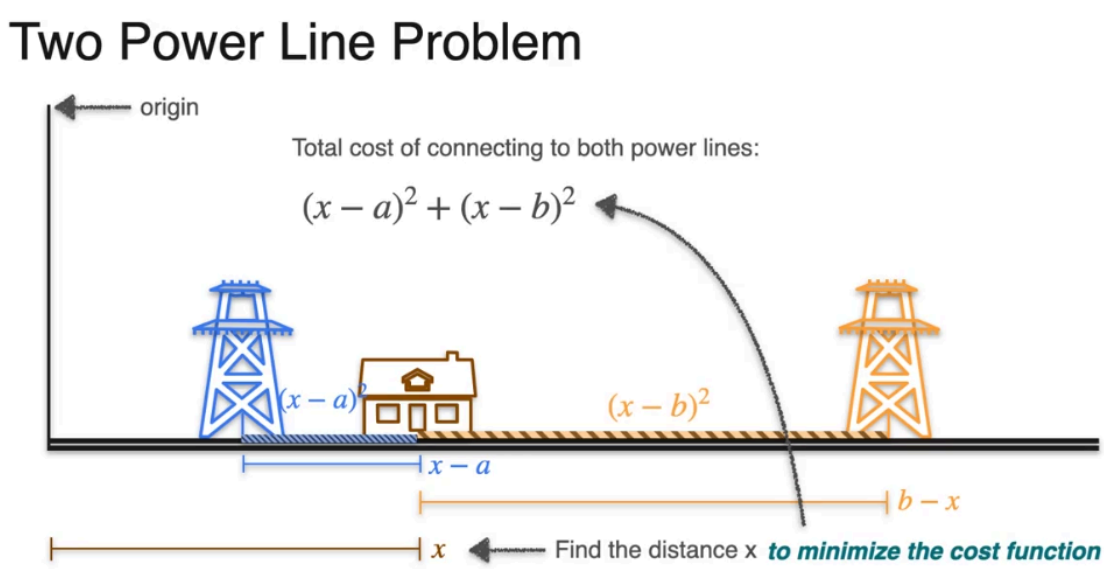


X.3. Optimization of squared loss - The two powerline problem

Goal is to minimize the cost function -> find x so it is small as possible.

In this example we use quadratic function to represent the distance.





So spoiler alert, the solution is to put it in the very middle of the two power lines to reduce the cost, and you're going to see that very soon.

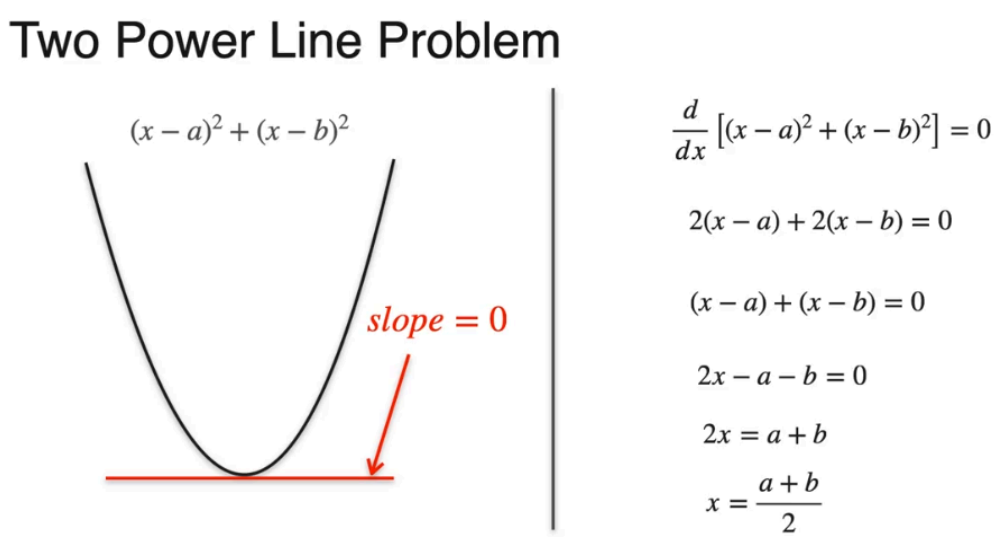
This is a **quadratic function**, and quadratics have a familiar shape: they **either open upwards or downwards**. In this case, it **opens upwards** because the **coefficient of x2 is positive**.

Now, in this **parabola**, we want to find the **minimum point**—the point where the **cost is lowest**.

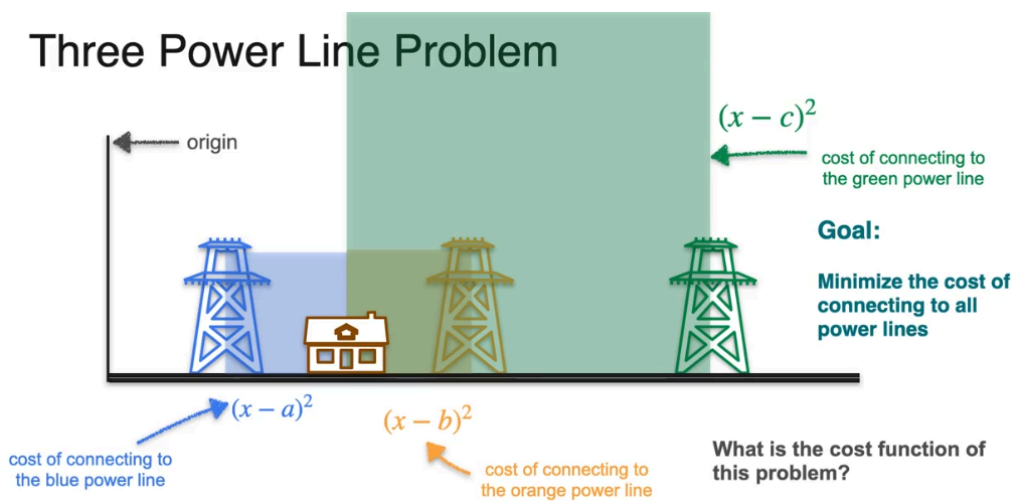
How do we do that? By finding the point where the **derivative is zero**.

At that point:

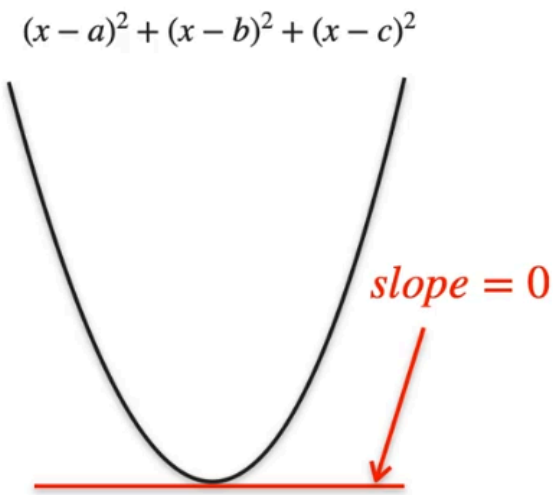
- The **slope of the tangent** is zero
- The tangent line is **horizontal**



X.3. Optimization of squared loss - The Three powerline problem



Cost function = Cost function = $(x-a)^2 + (x-b)^2 + (x-c)^2$



$$\frac{d}{dx}\left[(x-a)^2 + (x-b)^2 + (x-c)^2\right] = 0$$

$$2(x-a) + 2(x-b) + 2(x-c) = 0$$

$$(x-a) + (x-b) + (x-c) = 0$$

$$3x - a - b - c = 0$$

$$3x = a + b + c$$

$$x = \frac{a+b+c}{3}$$

Generalization :

The square loss is a very important loss function in machine learning.

The Square Loss

$$\text{Minimize } (x-a_1)^2 + (x-a_2)^2 + \cdots + (x-a_n)^2$$

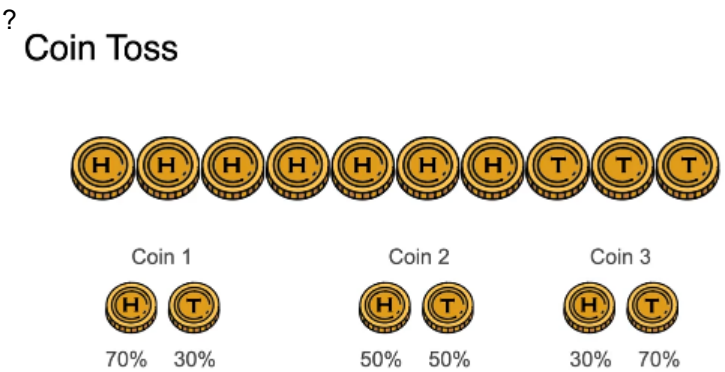
$$\text{Solution: } x = \frac{a_1 + a_2 + \cdots + a_n}{n}$$

XI. Optimization of log-loss -

XI.1.Part 1

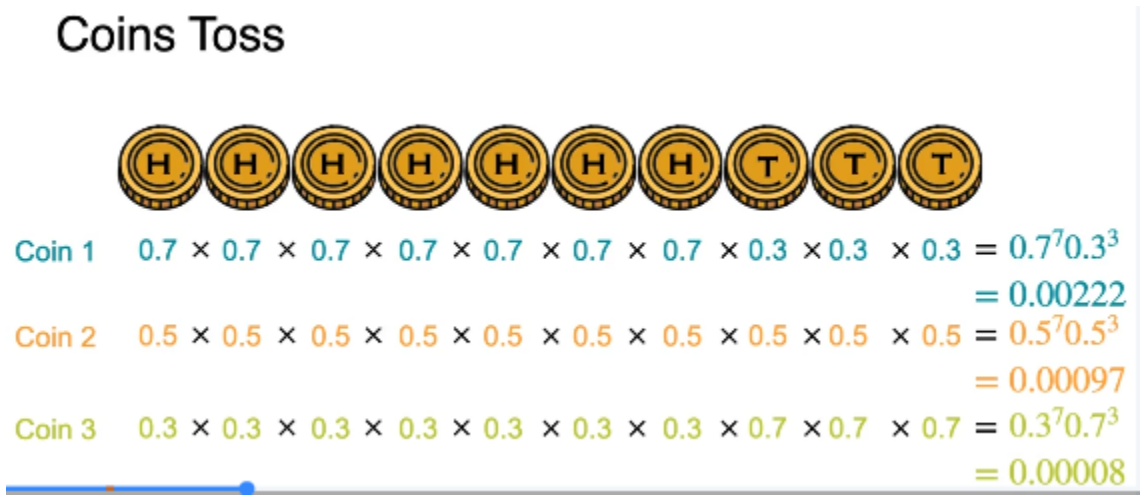
To win a game, you must have 7 heads and 3 tales.

Which coin do you choose to maximize your chances ?

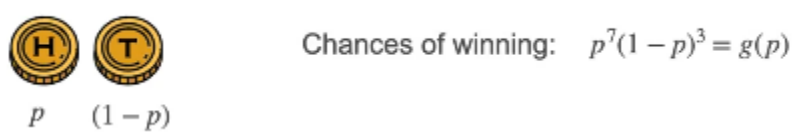


Probability for each case :

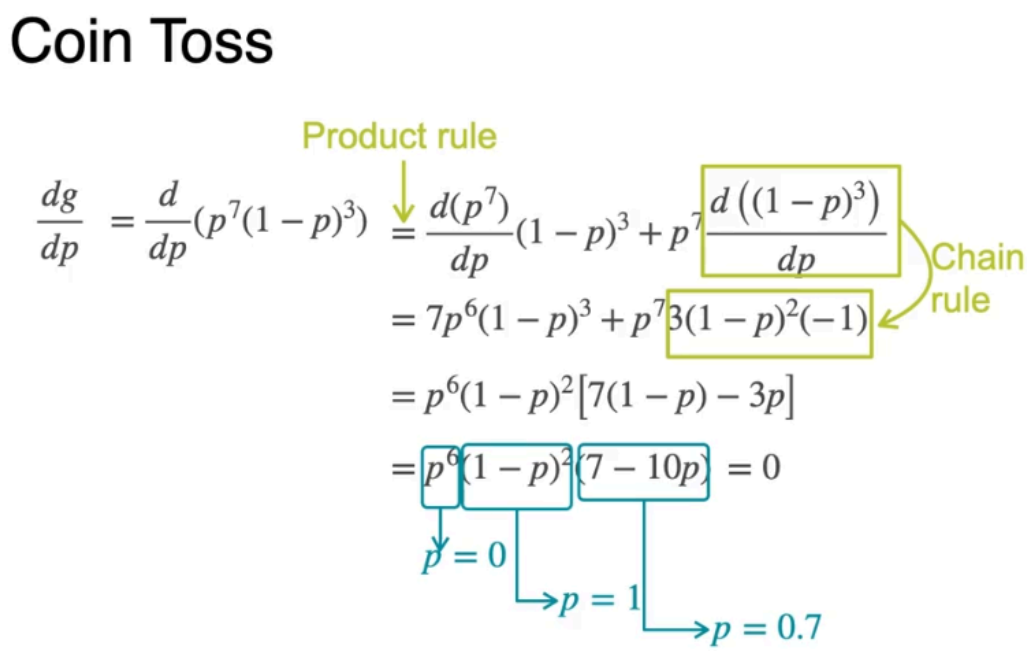
Yes the first is the best coin type to choose among those . Is this coin is best type for this case of all time?



Is it yes best choice of all time for this case.

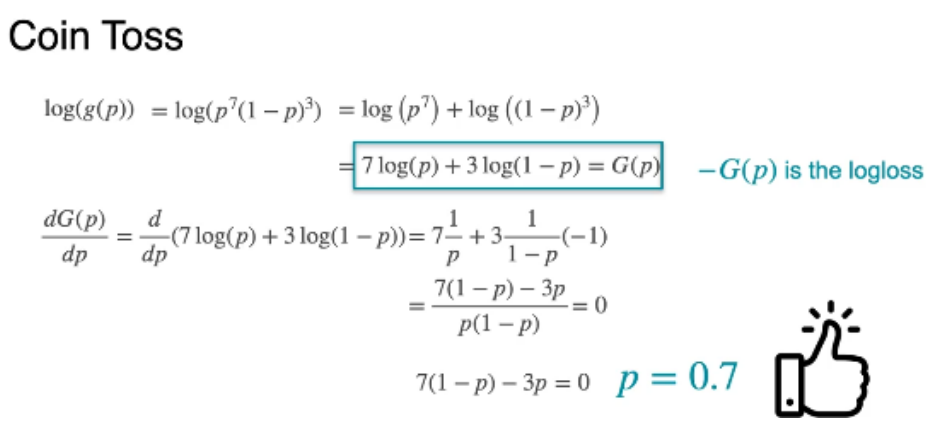


g(p) = p7 * (1-P)^3



Alot of work above, there is a solution.

Solution : by using log, maximizing log(g(x)) implies maximizing g(x).



Now, taking a **logarithm** is a very common trick in **machine learning**, especially when working with **probabilities**.

In fact, what we often use is the **negative of g(p)**, which is called the **log loss**. This is a **widely used loss function**, particularly in **classification problems**.

Why the negative? Because:

- The **logarithm of a probability** p (where0<p<1) is **negative**
- By taking **negative g(p)** we get a **positive number**
- Instead of **maximizing** g(p)g(p)g(p), we **minimize negative g(p)g(p)g(p)**

This simple trick makes optimization **easier and more consistent**, which is why log loss is so popular in machine learning.

XI.2.Part 2 - relation with ML

In the previous video, you found the **optimal coin** to win at a game.

If you look closely, what we did was **exactly like machine learning**.

- The **dataset** was the **10 coin flips**: seven heads followed by three tails.
- The **model** was the **coin itself**—a coin that lands on heads with probability PPP and tails with probability 1−P

What you did was **find the model (the best P) that most likely fits the dataset**.

How do you find that model? You **minimize the log loss**, and it turns out the **optimal P is 0.7**.

Relationship With ML

$p \quad (1-p)$

Data

Model

Minimized log-loss

$p = 0.7$

But why does the log loss use a **logarithm**? Why not just take the derivative of a big **product** of probabilities?

1. The reason is simple: **derivatives of products are hard**, while **derivatives of sums are easy**.

- Sure, you can use the **product rule** for two factors.
- But what about **three, four, or hundreds of factors**? It quickly becomes messy.

By taking the **logarithm**, you turn a **product into a sum**, making **calculation and optimization much simpler**. This is why **logarithms are so common in machine learning**—they turn messy products into manageable sums.

Why the Logarithm?

1. Derivative of products is hard, derivative of sums is easy

$$f(p) = p^6(1-p)^2(3-p)^9(p-4)^{13}(10-p)^{500}$$

$$\frac{df}{dp}$$

$$\begin{aligned}
&[6p^5](1-p)^2(3-p)^9(p-4)^{13}(10-p)^{500}+\\
&p^6\,[2(1-p)](3-p)^9(p-4)^{13}(10-p)^{500}(-1)+\\
&p^6(1-p)^2[9(3-p)^8](p-4)^{13}(10-p)^{500}(-1)+\\
&p^6(1-p)^2(3-p)^9[13(p-4)^{12}](10-p)^{500}+\\
&p^6(1-p)^2(3-p)^9(p-4)^{13}[500(10-p)^{499}](-1)
\end{aligned}$$

$$\frac{d}{dp}\log(f)$$

$$\frac{6}{p}+\frac{2}{1-p}(-1)+\frac{9}{3-p}(-1)+\frac{13}{p-4}+\frac{500}{10-p}(-1)$$

2. Another big reason is that **products of tiny numbers become extremely small**.

Imagine you have **1,000 probabilities**, each between 0 and 1. Multiplying them gives a **very tiny number**, so small that a computer might **not be able to handle it**.

If instead you take the **logarithm**, a very small number becomes a **large negative number**, which computers can handle easily.